

First Tragedy, then Parse: History Repeats Itself in the New Era of Large Language Models

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Abstract

Many NLP researchers are experiencing an existential crisis triggered by the astonishing success of ChatGPT and other systems based on large language models (LLMs). After such a disruptive change to our understanding of the field, what is left to do? Taking a historical lens, we look for guidance from the first era of LLMs, which began in 2005 with large n -gram models for machine translation (MT). We identify durable lessons from the first era, and more importantly, we identify evergreen problems where NLP researchers can continue to make meaningful contributions in areas where LLMs are ascendant. We argue that disparities in scale are transient and researchers can work to reduce them; that data, rather than hardware, is still a bottleneck for many applications; that meaningful realistic evaluation is still an open problem; and that there is still room for speculative approaches.

1 Introduction

Picture this scene: A renowned NLP researcher at a hot seven-year-old startup steps onstage to deliver a keynote. The speaker describes an ambitious new system to the packed room, building up to the results slide: a bar chart in which the x -axis shows the number of training words, and the y -axis shows system accuracy. As each data point is revealed, performance rises relentlessly, culminating in a system trained on well over a trillion words using over a thousand processor cores. It smashes the state of the art by a margin previously thought impossible.

Attendees are visibly shaken as they realize, over the course of a minute, that years of research have just been rendered utterly inconsequential. Established academics panic, anticipating the wholesale rejection of already-submitted grant applications. PhD students despair, contemplating the irrelevance of their unfinished dissertations. Many ponder an exit to industry or a change of fields. They will speak of little else this week.

More data is better data...

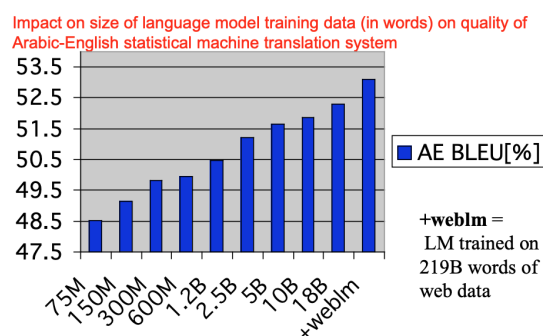


Figure 1: Results slide (reproduced from Och, 2005) of Franz Och’s keynote talk at the 2005 ACL Workshop on Building and Using Parallel Texts, a predecessor to the Conference on Machine Translation.

Does this scene sound like one that might have happened in the past year? In fact, it happened 19 years ago, in 2005, launching the first era of Large Language Models (LLMs): the **Statistical Machine Translation (SMT) era**. The speaker, Franz Och, had co-invented key methods in SMT (Och and Ney, 2003; Koehn et al., 2003; Och, 2003), but had not published new work since joining Google in 2004, instead revealing it in an invited talk prior to the launch of a new Google Translate (Och, 2006).¹ The provocative results slide from that talk (Figure 1) shows how Google improved its SMT system simply by expanding the training corpus of a phrase-based language model (Brants et al., 2007).²

¹The description of the talk and its aftermath is based on the vivid recollections of one of the authors, who was present.

²By **language model (LM)**, we mean a generative probabilistic model $\Pr(\mathbf{x})$ of a string $\mathbf{x}$. MT requires a *conditional* LM $\Pr(\mathbf{y} | \mathbf{x})$ of target string $\mathbf{y}$ given source string $\mathbf{x}$. In SMT it was originally modeled using the *noisy channel* formulation as $\Pr(\mathbf{y} | \mathbf{x}) \propto \Pr(\mathbf{y}) \times \Pr(\mathbf{x} | \mathbf{y})$ (Brown et al., 1993). The *translation model* $\Pr(\mathbf{x} | \mathbf{y})$ must be trained on a corpus of example translations, but the LM $\Pr(\mathbf{y})$ can be trained on *any* data in the target language, making it amenable to scaling. Like modern LLMs, LMs of the SMT era were generative probabilistic models, albeit based on n -grams (Shannon,

The first era of LLMs initially provoked great anxiety among MT researchers about the state of their field, but MT research has continued to flourish in academia, industry, and government. Even in the modern era of deep learning, MT has been a locus of innovations that have fundamentally altered NLP and all of machine learning (Bahdanau et al., 2015; Vaswani et al., 2017; Sutskever et al., 2014).

We believe that this history offers lessons for the *current* era of LLMs, an era during which massive proprietary models have become a de facto baseline for many tasks (Rogers, 2023). The expense of state-of-the-art research has led many to question the role of smaller and publicly funded groups in AI (Lee et al., 2023), a phenomenon we will call the **scale crisis**.³ Researchers without direct access to LLMs have publicly fretted over their research directions, with Togelius and Yannakakis (2023) suggesting pivots in research direction to sidestep scale, and Ignat et al. (2023) sketching research areas that are “not within the purview of LLMs.” But what should researchers do if they care about problems that *are* within the purview of LLMs? To answer this question, we look to the first era of LLMs. What were the durable lessons of that time and evergreen research problems that still matter today? We arrive at several recurrent lessons:

Scale is supreme (Section 2). We argue that, for areas where data is plentiful, NLP researchers cannot escape the **Bitter Lesson** (Sutton, 2019) that general purpose methods exploiting scale will outperform methods that leverage informed priors. We recommend that researchers take advantage of improvements in hardware as they enable scale at affordable budgets (Section 2.1) and that they remember small-scale problems (Section 2.2).

Evaluation is a bottleneck (Section 3). The Bitter Lesson favors generic methods, which require evaluation metrics to optimize over. But improved models create an evaluation bottleneck, since error detection becomes harder when most remaining mistakes are subtle or associated with

1948) rather than neural networks. Early LMs were widely used across applications, beginning in speech recognition (Jelinek et al., 1975), though, unlike contemporary LLMs, they were rarely end products themselves. Although the LM of Brants et al. (2007) had a very different architecture from contemporary LLMs, it was an LLM in an important sense: it was trained on 2 trillion tokens, which is comparable to the training data size of modern LLMs.

³We use the term *crisis* deliberately since others have done so. For example, following the announcement of GPT-4 (OpenAI, 2023), @andriy_mulyar (2023) posted on Twitter that his feed was “full of ph.d. students having an existential crisis.”

edge cases. At scale, automated metrics show their flaws. We recommend that researchers work on improving metrics (Section 3.1).

There is no gold standard (Section 4). When one can afford the annotation costs, it may be tempting to consider human feedback as the ideal solution to the evaluation problem. Unfortunately, history has repeatedly shown that naïve methods of soliciting human preferences result in poor feedback, prioritizing superficial properties of model outputs. This lesson may serve as a counterpoint to the impulse to collect massive quantities of low-quality data in response to the Bitter Lesson. Instead, we recommend grounding performance measurement in concrete downstream tasks (Section 4.1).

Progress is not continuous (Section 5). The ascent of neural MT abruptly ended SMT’s decade of seemingly unbeatable growth. This change in directions was enabled by new hardware-based paradigms, so we recommend that researchers continue exploring new methods that might scale well on future hardware (Section 5.1).

We conclude with a simple message: **Do research** (Section 6). We remind the reader that engineering achievements do not render scientific achievements insignificant, and we encourage the NLP community to renew their commitment to foundational scientific research even in areas where scale is currently a dominant factor.

2 Scale is supreme.

The first lesson offered by the history of SMT is that data and compute scale are the dominant factors in system performance. In all eras of MT, improvements in BLEU are logarithmic in training data size (Brants et al., 2007; Koehn and Knowles, 2017). This is immediately obvious from Figure 1: in order to achieve each linear step of improvement in accuracy (y axis), training data size must double (x axis). Figure 2 (reproduced from Kaplan et al., 2020) shows a strikingly similar log-linear relationship between training data size and system performance for LLMs. Indeed, such relationships are observed across many application areas of machine learning, including vision (Mahajan et al., 2018) and speech (Moore, 2003).

In a research landscape centered on performance metrics, scale will dominate. Sutton (2019) named the resulting malaise the “Bitter Lesson”: “General methods that leverage computation are ultimately the most effective, and by a large margin.” Both

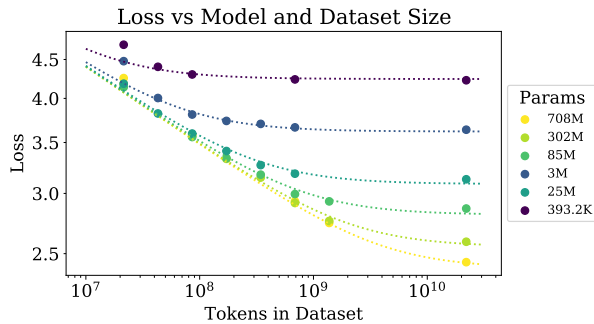


Figure 2: Figure from Kaplan et al. (2020) illustrating a power law relationship between dataset size and test loss for LLMs with varying numbers of parameters.

SMT and LLMs exemplify this lesson, and in fact Sutton explicitly references natural language processing. Many NLP researchers accordingly feel lost without access to large-scale systems. However, there are limits to scale, and as we will show by reviewing the history of the SMT era, its disparities are often transient. **The scale crisis is not a permanent state.**

2.1 Follow the hardware.

For several years following the release of Google Translate, large-scale commercial systems dominated the rankings of translation into English, where data was plentiful (NIST, 2008; Callison-Burch et al., 2009, 2010, 2011, 2012). Open source tools such as Moses (Koehn et al., 2007) and collaborations that pooled the resources of small labs narrowed the gap but did not close it until 2013, when translation into English was convincingly won by an academic group using modest hardware (Bojar et al., 2013). The decisive tool was KenLM, an efficient language modeling library (Heafield, 2011; Heafield et al., 2013) that demonstrated how, with the right software, contemporary hardware had made LLM training widely accessible. This end to the SMT scale crisis was the outcome of trends in hardware and software advancement.

The advent of LLMs in the SMT era and their later academic availability were both consequences of **Moore’s law**, a six-decade trend in which computing power has doubled biannually: as Sutton (2019) observes, “over a slightly longer time than a typical research project, massively more computation inevitably becomes available.” SMT-era LLMs arrived when researchers noticed that they had ignored Moore’s law for too long. They rapidly closed the gap: Brants et al. (2007) ended a brief race to scale n -gram models to web-scale

data (Zhang et al., 2006; Emami et al., 2007). But once the gap was closed, further incremental improvements—which required doubling the training data—necessarily required doubling the hardware cost or waiting for its capacity to double. Well-funded research sought more immediate gains elsewhere, while researchers with longer horizons rode Moore’s law towards parity through collaboration and algorithmic advances. The new era of LLMs has already followed the first part of this pattern: computational requirements of LLMs have been doubling at a rate of less than a year or perhaps faster (Sevilla et al., 2022; Amodei and Hernandez, May 16, 2018), much faster than Moore’s law.

The end of the SMT scale crisis was by no means inevitable or foreseeable at the beginning of the first era of LLMs in 2005: it resulted from the efforts of many researchers. We are encouraged to see a similar trajectory forming now. Already, startups advertise cheap large-scale training to the public (Portes et al., 2023). Like the groups that competed with commercial translation software in the SMT era, large cross-institutional collaborations are currently pooling resources to build public models (Scao et al., 2022). A community has developed around efficient ML, spawning new publication venues like MLSys and developing algorithms already employed in many LLMs (Hernandez and Brown, 2023). BERT (Devlin et al., 2019), regarded as inaccessible to many small academic labs at its release, now runs on a consumer-grade M1 MacBook laptop (Roesch and Mazonett, 2021).

In short, small labs do not need to abandon their entire research direction if they are interested in working with state-of-the-art models. Algorithmic efficiency guarantees usually hold across different resource scales, so a method developed on inexpensive hardware can be directly applied at industrial scale. Therefore, *all researchers can seek opportunities to collaborate and develop better algorithms.*

2.2 Remember small-scale problems.

While directly tackling scale is one strategy, we also recommend pursuing research on problems where data, not compute, is the bottleneck.

Small-scale settings provide a fertile ground for innovation in data-driven methods; in the previous era of LLMs, SMT researchers often used linguistic structure to improve performance when using smaller data. When such methods showed promise in these development settings, Google Translate inevitably tested them at industrial scale. Nonethe-

less, at the end of its lifespan, Google Translate’s SMT system remained a phrase-based lookup table. The exploration of classical SMT researchers led to no lasting inventions based on syntax or semantics, and so the assumption that resulting improvements could transfer to large scale settings may have been based on unfounded optimism. Unlike their predecessors, modern NLP researchers preemptively recognize the futility of scaling up data-informed methods, because many of the improvements they offer are already provided by scale. But by leaning too far into the bitter lesson’s pessimism now, we risk neglecting settings where, for practical or financial reasons, we *must* learn from limited data.

When Google Translate launched the first era of LLMs, it was only available between Arabic and English (Och, 2006). Data was the bottleneck that stood between SMT and its application to many meaningful problems, including, most obviously, the translation between many other language pairs. Solving this bottleneck required such diverse efforts as the collection of the Europarl corpus (Koehn, 2005), the OPUS corpus (Tiedemann and Thottingal, 2020), the JHU Bible Corpus (McCarthy et al., 2020) and the Nunavut Hansard (Martin et al., 2003); the rapid development of a Haitian Creole corpus in the aftermath of the Haiti earthquake (Lewis, 2010); the crowdsourcing of corpora for many Arabic dialects (Zbib et al., 2012); and the development of open-source web crawlers for parallel text (Smith et al., 2013). It is ongoing today in projects led by groups like Masakhane (Adelani et al., 2022; Nekoto et al., 2020; Emezue and Dos-sou, 2021) and No Language Left Behind (NLLB team et al., 2022). And yet, MT is still impossible for the vast majority of the world’s estimated 7,000 languages.

Just as no one would have claimed in 2006 that Google had solved all translation problems, no one should claim now that LLMs have solved all NLP problems.⁴ An identical bottleneck persists in the current era of LLMs, dominated by anglophone systems like ChatGPT. While these models can handle many languages to some degree due to the

incidental multilinguality of any large training corpus (Blevins and Zettlemoyer, 2022), the training data is overwhelmingly English, and supervision data for learning from human feedback is overwhelmingly from English-speaking Kenyans (Per-rigo, 2023). The hegemony of English has made it a presumed default, inciting the creation of the Bender Rule: “Always name the language(s) you’re working on” (Bender, 2019).

Just as MT researchers have done since the SMT era, LM researchers today develop tools which rely less on scale (Alabi et al., 2022; Meyer et al., 2022; Park et al., 2021) for underserved languages. Researchers can leverage international collaborations with local linguists, incentives outside a profit model, and noncommercial resources to broaden the population that has access to technology in their own language. Furthermore, as compute costs continue to decline, even English corpora will become relatively “low resource” for future highly overparameterized models. To exploit these datasets more effectively, we turn to a perennial need across AI: quality evaluation metrics.

3 Evaluation is a bottleneck.

The next lesson offered by reflection on the SMT scale era is that the quality of evaluation methods makes a substantial difference in the effectiveness of training because a good evaluation can be used as a training signal. In SMT, this epiphany was delivered by minimum error rate training, which trained directly on target metrics like BLEU (Och, 2003). Likewise, train-time feedback metrics are often adapted for test-time evaluation: language modeling work may present validation loss, or equivalently perplexity, as the direct measurement of language modeling performance. It is therefore easy to use **symmetric** evaluations, applying the same metric for training feedback and test-time performance assessment. Like model evaluation, training can be based on comparison with a ground truth, as in conventional training; quality estimation based on output alone, as used often in Reinforcement Learning (RL) settings (Konda and Tsitsiklis, 1999; Silver et al., 2014; Bai et al., 2022b); or direct feedback, as provided by RL from Human Feedback (RLHF) and related methods of human assessment (Ziegler et al., 2019; Stiennon et al., 2020; Ouyang et al., 2022; Bai et al., 2022a). The evaluation metric then becomes a crucial lever to improve model quality.

⁴While we focus on data scarcity in underserved languages, some settings may provide limited data even in English. Such data scarcity may be due to practical hurdles to data collection (e.g., legally protected medical data) or an insufficient profit incentive (e.g. data for speakers of English from lower socioeconomic classes; Curry et al., 2024) While we cannot say the degree to which pure scale can solve problems like robustness or handling longer contexts, we can identify many problems where data is not collected at scale.

With large scientific projects in both industry (StabilityAI, 2021; Thoppilan et al., 2022) and non-profit sectors (Biderman et al., 2023; Scao et al., 2022) spending millions on training LLMs, it may be surprising to point away from scale to other bottlenecks in system performance. The focus on scale bottlenecks is reasonable, as for any given compute budget, there is an optimal quantity of training data that yields the most accurate model (Hoffmann et al., 2022). Either compute or data size, therefore, can become a limiting factor, and there are still financial barriers to compute (Lee et al., 2023) and limits on how much unstructured natural language data is available (Villalobos et al., 2022). However, training requires data and compute resources to be connected by evaluation feedback; this connection determines the effectiveness of training.

Unfortunately for efficient automatic approaches to evaluation, automatic metrics often fail to predict human evaluation, as shown for language modeling loss (Liu et al., 2023) and BLEU (Reiter, 2018). Even a metric that mirrors the training objective slightly can artificially inflate model performance; for example, it was observed early on that a phrase-based evaluation metric like BLEU might favor phrase-based SMT (Riezler and Maxwell III, 2005). Identifying good metrics is challenging and becomes more difficult with each improvement, as the remaining errors become increasingly subtle or complex. This challenge was also recognized early in SMT, with calls for a “BLEU++” (Och, 2005).

Evaluation, therefore, has become a crucial goal for modern LLM research. While training often relies on cross-entropy loss or other simple comparisons between each token in a sequence, benchmarking a trained LLM typically uses different evaluation criteria, such as checking the final answer in a word problem or the accuracy of a prompt-based classifier (Laskar et al., 2023). However, these tests are plagued by data contamination: benchmark exposure during training has created illusory gains in tasks ranging from code generation (Khan et al., 2023) to theory-of-mind puzzles (Ullman, 2023). Clearly language models are improving, but we cannot say precisely how, or by how much.

3.1 Improve the metrics.

Because evaluation is a bottleneck, we recommend greater focus on improving metrics, a goal which can yield rewards even with limited access to scale. Straightforward increases in computational infras-

tructure and raw data collection yield predictably diminishing returns on investment. Evaluation, by contrast, provides a conceptual space that can reward innovation and careful work with new insights and unknown improvements in system capabilities. What are the fundamental problems in this space, and how might we approach them?

One argument for why automatic metrics and static benchmarks are poor methods of evaluation is that they fail at measurement modeling; that is, these metrics do not actually measure what they purport to measure. This concern is reflected in objections to benchmarks and metrics that fail to reflect human evaluation (Liu et al., 2016; Novikova et al., 2017) or improvement on natural language understanding more broadly (Raji et al., 2021). These discussions parallel the unease in the MT community when studies found that automated metrics such as BLEU did not always correlate with human judgments (Callison-Burch et al., 2006). In response, SMT saw a flurry of incrementally improved bitext-based metrics (Stanojević and Sima'an, 2014; Popović, 2015; Mutton et al., 2007). In a scale crisis, improving evaluation metrics that leverage naturally available data like bitext can be a worthwhile focus.

Unfortunately, the naturally available data used in evaluation can still contaminate training corpora. Furthermore, automated metrics that rely on a static ground truth cannot reflect general quality (Raji et al., 2021), model conditions under interactive deployment, or provide on-policy reward feedback for reinforcement learning. These issues motivate automated metrics that do not require ground truth, although proposals based on AI supervision are themselves difficult to evaluate due to the same issues of dataset bias and contamination.

Despite a research community strongly motivated to improve them, even the best automated metrics are far from perfect. As automated metrics and static benchmarks fail, researchers with resources are increasingly hiring humans to assess model outputs. Facing similar challenges in SMT, researchers also called for human evaluation to be prioritized, both for benchmarking (Callison-Burch et al., 2006, 2007) and for training (Hopkins and May, 2011). However, human evaluation does not intrinsically solve problems with measurement modeling, and raises challenges of its own.

4 There is no gold standard.

Language evolved to be interpreted by humans. This fact leads us to a tempting myth: that we can easily evaluate synthetic natural language outputs by simply asking a human for their opinion. To the contrary, the next lesson we discuss from the SMT era is that human annotation cannot provide a universal “gold standard” of quality feedback. When MT competitions proudly turned to human evaluation as the highest-quality and most reliable option for choosing a winner, critics pointed out that these evaluations failed the basic expectations of consistency needed for a fair ranking (Bojar et al., 2011; Lopez, 2012). Even soliciting useful, let alone perfect, evaluations from humans turned out to require careful thought and trade-off decisions.

Practitioners often rely on naive methods of soliciting human feedback on LLM outputs, such as single rating scales or ranking model outputs by quality, which do not distinguish why annotators prefer a particular model output and thus offers limited guidance. For example, OpenAI’s ChatGPT annotator interface asks the user to rank outputs from “best” to “worst” (Ouyang et al., 2022). Anthropic adds extra dimensions but with limited guidance, as annotators evaluate the extent to which generated text is “helpful” and “harmful,” claiming that the vagueness of these guidelines permits versatile human preferences (Bai et al., 2022a). Recent work goes even further by using freeform text feedback, rather than predefined numerical axes (Shuster et al., 2022; Andreas et al., 2022; Scheurer et al., 2022, 2023), although how best to incorporate these explanations remains an unsolved problem.

The NLP community, however, is rediscovering that eliciting human preferences without clear guidance produces data that is not only noisy, but introduces systematic errors in models trained on the data. When many dimensions of quality are collapsed into a single preference scale, outputs that are worse along some dimensions may have higher ratings because they perform well along others. In these cases, annotators prioritize fluency over other aspects of the text, such as factuality or consistency (Clark et al., 2021). LLMs consequently prioritize fluency of large language model outputs over factuality (Ji et al., 2023), mirroring concerns from the SMT era that models prioritized fluency over the faithfulness of translations (Dorr et al., 2011)—concerns that have since been empir-

ically confirmed (Martindale and Carpuat, 2018). Belz and Hastie (2014) and van der Lee et al. (2021) note that overall quality of generated text is often “too abstract” to be measured and both recommend the use of separate criteria for different dimensions of the text to distinguish what specific issues are present in a model output. Gehrmann et al. (2023) and van der Lee et al. (2021) warn that vague annotation guidelines can exacerbate annotator confusion, underscoring the importance of clearly defining the different dimensions on which to rate text quality.

Even after these refinements to the evaluation process, human evaluation for MT has encountered issues that remain unsolved, and current research suggests that evaluation of current models will increasingly encounter similar issues, including the following challenges.

Specifying evaluation criteria is hard. Even when evaluation criteria are separated into several axes, these scores are correlated, suggesting that human evaluators have difficulty in separating out criteria such as adequacy and fluency (Novikova et al., 2018). In addition, many studies fail to define their axes (van der Lee et al., 2021), permitting evaluators to differ even more in their interpretation of the task and thus increasing variation among annotators. That is, separating out axes of evaluation is necessary but not sufficient to identify multiple desirable traits of model output, a recurring problem in the history of MT evaluation (Chatzikoumi, 2019). Even when evaluation criteria can be defined clearly, crowdsourced annotators often lack the necessary expertise to follow them. Crowdworkers therefore align poorly with expert annotators, even underperforming against automatic evaluation metrics (Freitag et al., 2021). Some problems with objective specification can be resolved by defining multiple objectives and consulting expert annotators rather than crowdworkers.

Individual preferences are inconsistent. Classic SMT results reveal another fundamental problem in human evaluation: pairwise human rankings often fail to produce a consistent order (Bojar et al., 2011; Lopez, 2012). Any approach based on comparing outputs therefore reflects an unrealistic expectation of consistency in human preferences. The signal provided by ranking is noisy.

Disagreement isn’t just noise. When human evaluators disagree on the quality of text, this

does not necessarily reflect “noise” or “random variation” but rather genuine differences in opinion among evaluators (Larimore et al., 2021; Patton et al., 2019; Prabhakaran et al., 2021; Pavlick and Kwiatkowski, 2019; Basile et al., 2021; Plank, 2022), a problem that has long plagued MT evaluation (Lommel et al., 2014). The management of diverse annotator preferences is only exacerbated when benchmarking and training on freeform text from varied sources (Giulianelli et al., 2023). Furthermore, aggregation of annotator judgments obscures the opinions of underrepresented groups (Prabhakaran et al., 2021; Fleisig et al., 2023), and use of inter-annotator agreement as a quality metric causes additional erasure of perspectives by denying that these priorities are contested (Blodgett, 2021).

These issues collectively prevent human evaluation from providing clear feedback on model outputs. The fact that they have remained major concerns in MT despite decades of research suggests that current researchers would do well not to underestimate the challenges posed by these issues. Furthermore, issues of both task specification and disagreement may be even more central to the evaluation of current models that can handle more varied tasks. Whereas fluency and faithfulness to a source text might cover major concerns in MT, there is a broader range of criteria that generated text must fulfill, such as informativeness and coherence (van der Lee et al., 2021). These requirements, along with increased freedom to produce text on topics where there is real-world disagreement, including social, ethical, and political concerns (Abid et al., 2021; Blodgett et al., 2020; Liu et al., 2021; Zhao et al., 2021), mean that human evaluation issues will be pressing problems for the NLP community to solve.

4.1 Focus on concrete tasks.

Due to the inherent flaws of evaluation based on human assessment, we recommend measuring concrete tasks under deployment conditions. Extrinsic evaluations (Belz and Reiter, 2006), wherein model output quality is evaluated based on utility for specific downstream applications, are still uncommon in evaluation of text generation (van der Lee et al., 2021). However, they may be more useful for evaluating the quality of content or meaning (Reiter and Belz, 2009; Reiter, 2023) because human assessment often fails to predict performance

on downstream applications (Kunz et al., 2022).

In MT, concrete downstream objectives have long been used in evaluation. Snover et al. (2006) examined how many manual edits human translators had to make to model output, reflecting the desiderata of human-AI collaboration settings. Other metrics rely on the user’s ability to accomplish specific tasks using model output, such as answering reading comprehension questions based on translations (Jones et al., 2005; Callison-Burch, 2009; Scarton and Specia, 2016) or summaries (Wang et al., 2020). A recent and growing body of research attempts to measure the effectiveness of MT in second language education (Lee, 2023).

In general, the best evaluations are likely to rely on realistic assessment of what LLMs enable humans to do. In modern LLMs, work on the challenges of evaluation is likely to draw on insights from human-computer interaction.⁵ Good user trials require careful study design and consideration of human variety, as well as an understanding of individual psychology.

5 Progress is not continuous.

Our final lesson is that new paradigms can unlock new orders of scale and even new scaling coefficients, leading to abrupt improvements in performance. The SMT era, fueled by large n -gram models, lasted for over a decade, with scale providing increasing improvements over time. But Moore’s law was threatened during this era due to the breakdown of Dennard scaling, the observation that smaller transistors require commensurately less power, meaning that they can be miniaturized while keeping power consumption constant. To drive continued improvement, hardware manufacturers turned to parallelization. Graphical processing units (GPUs), which favor high parallelization of code with minimal branching—and thus simpler and smaller processors—were soon being repurposed to train neural networks (Hooker, 2020).

Neural networks had been investigated in SMT for years. Indeed their earliest use in SMT was as n -gram language models, when Schwenk et al. (2006) built an SMT decoder using the neural n -gram model of Bengio et al. (2003)—an idea that only began to gain traction almost a decade later

⁵Interdisciplinarity between machine learning applications and HCI is a perennial concern. HCI researchers are periodically invited to speak at ML and ML applications conferences, e.g., NeurIPS hosted HCI-centered keynotes from Deborah Estrin in 2013 and Juho Kim in 2023.

when revisited by [Devlin et al. \(2014\)](#), whose ACL best paper award marked a shift in NLP establishment attitudes towards neural networks. The increasing power of GPUs and their use in training neural networks fueled new research in end-to-end neural MT (NMT), enabling [Kalchbrenner and Blunsom \(2013\)](#) to revive the even older idea of an encoder-decoder architecture ([Ñeco and Forcada, 1997](#)). Advances such as attention ([Bahdanau et al., 2015](#)), seq2seq ([Sutskever et al., 2014](#)), and transformers ([Vaswani et al., 2017](#)) followed in a flurry of activity. Within two years, NMT swept the annual shared tasks ([Jean et al., 2015](#); [Chung et al., 2016](#); [Bojar et al., 2016](#)), and in 2016, Google Translate announced that it had switched to NMT ([Wu et al., 2016](#); [Turovsky, 2016](#)). Research on SMT quickly faded.

GPUs effectively introduced a new dominant **paradigm** by creating conditions that favored deep learning. [Kuhn \(1962\)](#) described scientific advancement as a cycle of scientific revolutions in which paradigms such as phrase-based SMT or deep learning emerge, followed by periods of **normal science** when researchers aim to apply, articulate, and expand the fact base of the paradigm. Often, scientific revolutions result from the availability of new tools. [Hooker \(2020\)](#) drew on this framework of scientific revolution to analyze the landscape of AI research, identifying the **Hardware Lottery** as a situation in which hardware dictates methods.⁶ Under the Hardware Lottery, GPUs offered a winning ticket for deep learning to reshape MT. What research objectives are recommended by the resulting NMT revolution?

5.1 Shape the hardware.

The Hardware Lottery tells us that hardware guides the direction of research, but researchers can also direct the design of hardware. While these new tools may enable scientific revolutions, [Kuhn \(1962\)](#) pointed out that the development of new tools is itself shaped by the reigning paradigm and by the normal scientific process. Hardware design itself is an example, having been driven for many years by incremental improvements to a paradigm of miniaturization and parallelization of transistors. But hardware manufacturers are approach-

ing the physical limits of miniaturization, and the path forward is again uncertain ([Lundstrom and Alam, 2022](#)), as it was at the end of Dennard scaling. Therefore, we recommend that researchers focus not only on developing and using new hardware, but on anticipating potential hardware developments and developing algorithms for platforms before they are widely available.

By creating software tools and algorithms that can take advantage of hardware designed for sparsity ([Krashinsky et al., 2020](#)) or new sources of parallelism ([Launay et al., 2020](#)), researchers can develop techniques preemptively for future technologies. At the same time, they also create a market to motivate the development of new hardware that can enable the next revolutionary development. Researchers may even co-design hardware and software jointly, a strategy likely to drive future computing advances ([Leiserson et al., 2020](#); [Lundstrom and Alam, 2022](#)). It is the possibility of reshaping tools for the future that makes alternative paradigms worth exploring under a scale crisis.

6 Conclusion: Do research.

As pure engineering efforts and institutional wealth outstrip novel scientific work, some in the AI community are pessimistic about the prospects of foundational research. Our position, articulated over the course of this paper, is that there is much exciting, timely work yet to be done.

These lessons are not particular to LLMs, but apply to any field subject to the Bitter Lesson. For example, issues in human evaluation plague many disciplines in machine learning. In computer vision, annotator idiosyncrasies account for many of the remaining inaccuracies of modern ImageNet models ([Shankar et al., 2020](#)). If a constrained labeling task such as image classification is subject to varied human judgment ([Parrish et al., 2023](#)), how much harder is it to annotate free text generation?

Beyond our specific recommendations for researchers interested in improving the capabilities of language models, we would also point to scientific opportunities across related fields. From interpretability to empirical training analysis to public policy, many research areas only become more relevant and complex as models rapidly improve. Furthermore, while novel modeling work suffers in a scale crisis, we can focus on new architectures and algorithms that take advantage of existing hardware and even anticipate future tools.

⁶[Gururaja et al. \(2023\)](#), whose oral history of NLP—including comments on the current scale crisis and the cyclic nature of what they call exploit-explore incentives—complements our work, also point to similar paradigm shifts in NLP emerging from a *software* lottery.

Some speculative alternatives to phrase-based SMT, such as explicitly modeling syntax (Galley et al., 2004; Collins et al., 2005; Chiang, 2007) and semantics (Jones et al., 2012), were obviated by the Bitter Lesson and its expression in NMT. However, other proposals formed the basis of the NMT era. Many enduring careers in NLP research were forged in areas that are now forgotten, and without risky exploration of unproven directions, the field could not have achieved many breakthroughs. Our anxieties should not discourage us from seizing the opportunities presented by a new era of LLMs.

Limitations

The positions taken in this paper are based on both experience and reading of historical trends in natural language processing. While we believe that the lessons we identify in this paper are durable, history does not always repeat, and our oracular powers are otherwise limited. Even after considering our position, researchers should use their own best judgement on directions to pursue.

Ethical Considerations

The authors received permission from Andriy Mulyar to feature his Twitter post as an example of the March 2023 scale crisis discourse.

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